

On the Age Calibration of Open Clusters using Red Clump Stars

ABIGAIL R. CHRISS ¹ AND GUY WORTHEY ²

¹*Department of Physics and Astronomy, Bowdoin College, 8800 College Station, Brunswick, ME 04011-8488, USA*

²*Department of Physics and Astronomy, Washington State University, 1245 Webster Hall, Pullman, WA 99164-2814, USA*

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ABSTRACT

In this study, we extend the dust-independent [Hatzidimitriou \(1991\)](#) relation between cluster age and d_{B-R} color difference between red giant branch (RGB) and red clump (RC) to younger cluster ages. We perform membership analysis on twenty-two open clusters using *Gaia* DR3 astrometry, then compute the difference in color of red giant branch and red clump d_{B-R} using *Gaia* photometry. We find that the trend derived from older clusters does not extrapolate to younger ages and becomes double-valued. We confirm that d_{B-R} is independent of metallicity. Current stellar evolutionary isochrones do not quantitatively reproduce the trend and furthermore predict an increased color gap with a decrease in metallicity that is not echoed in the data. Integrated light models based on current isochrones exaggerate the color change over the $-0.5 < [\text{Fe}/\text{H}] < 0$ interval at the few-percent level.

Keywords: Open star clusters (1160) — Stellar ages (1581) — Red giant clump (1370) — Red giant branch (1368) — Hertzsprung Russell diagram (725)

1. INTRODUCTION

Star clusters represent both a crucial testing ground for the theory of stellar evolution and markers for the chemical and dynamical history of the Milky Way and other galaxies. While massive star clusters typically host multiple stellar populations ([Gratton et al. 2012](#)), lower-mass open clusters are currently presumed to be simple stellar populations (SSPs) of a single age and heavy element abundance pattern. Cluster ages are often derived from the color magnitude diagram (CMD), where, especially, the luminosity of the main sequence turnoff (MSTO) provides a theoretically robust chronometer ([Chaboyer 1995](#)).

Even assuming perfect comparison models, distance uncertainty and line of sight dust extinction add error to age estimates. [Hatzidimitriou \(1991\)](#), hereafter [H91](#), proposed a method that bypasses both dust and distance. The color of the He-burning red clump (RC) is subtracted from the color of the H-burning red giant branch (RGB) at the same luminosity. [H91](#) used Johnson-Cousins B and R filters and thus the color difference is d_{B-R} . Age was found to track d_{B-R} for clusters older than ~ 2 Gyr and for a variety of heavy element abundances. A bonus advantage of this method is that red clump stars are ~ 100 times brighter than MSTO stars, and thus the method could be applied to distant clusters.

The advent of star formation history reconstruction methods, where the entire CMD is fit with a swarm of stellar evolutionary isochrones ([Harris & Zaritsky 2001](#); [Dolphin 2002](#)), might explain why the [H91](#) formula has not seen wide use ([Girardi 2016](#)). [Girardi](#) notes, however, that the reconstructions for older ages rest upon clump and RGB lifetimes, something predicted by stellar evolutionary theory only at the 20% level.

The purpose of the present work is to confirm and extend [H91](#)’s result. The red clump should exist in the CMD for ages as young as ~ 200 Myr, or MSTO masses of $\sim 5 M_{\odot}$. We were curious if [H91](#)’s result extended to younger populations. To that end, we mined *Gaia* open cluster data for additional clusters with red clumps, with particular attention to clusters between 200 Myr and 2 Gyr in age.

We describe cluster membership discrimination and our derivation of isochrone-based ages in §2. The [H91](#) method is addressed in §3, and we discuss the result in §4.

2. CLUSTER SAMPLE AND MEMBERSHIP ANALYSIS

From open cluster lists ([Kharchenko et al. 2013](#); [Cantat-Gaudin et al. 2020](#)) we selected clusters beyond the [H91](#) list, initially in the age range 200 Myr to 2 Gyr. As the project developed, we added a few older clusters to test the repeatability of the [H91](#) method. Ideal clusters lie nearby and bloom richly with stars.

For each of the 22 clusters presented in Table 1 we performed a membership analysis before applying reddening corrections and analyzing the CMD. The probabilities of membership for stars in each cluster were estimated using position (α, δ) , proper motion (μ_α, μ_δ) , and parallax (π) from *Gaia* Data Release 3 (DR3). Cluster centers, proper motion central locations, and parallaxes from Simbad served as a first guess for each cluster, though we allowed drifts from these values during the fitting process. For position and proper motion, we drew annuli around the central location and computed star density (number per square degree or number per km s^{-1} , respectively) as a function of annulus bin radius (r) . We employed a least squares fitter to model functions for the total stars and the field star background. We modeled the cluster spatial profile as a Gaussian with a constant-density pedestal representing field stars (Fig. 1). Based on position, for example, the cluster membership probability for each star i is

$$P_{\text{cluster}, (\alpha, \delta)}(i) = \frac{N_{\text{total}}(r_i) - N_{\text{field}}(r_i)}{N_{\text{total}}(r_i)},$$

where r_i is the distance of a star from the cluster center, and the number of counts per square degree N come from two (cluster and field) function fits to the density profile. In other words, the probability is the expected fraction of member stars as a function of distance from the cluster center.

The fit for proper motion assumed an exponential density profile for the cluster on top of a constant field density (Fig. 2), the only difference being that the “distance” is the $\Delta\mu$ from the cluster locus. The parallax distribution was fit with two Gaussians, a broad one for the field and a narrow one for the cluster (Fig. 3). Cone search opening angles (column 6 in Table 1) were revisited if density profiles did not clearly reach an asymptote.

Given three separate probability estimates, the combined probability from separate $P_{(\alpha, \delta)}$, P_μ , and P_π estimates is computed as in Balaguer-Núñez et al. (1998), updated to include increased dimensionality as in Griggio & Bedin (2022). The resultant probability distribution for NGC 7789 is shown in Fig. 4.

3. RESULTS

From cleaned CMDs such as that of Fig. 5, we computed clump color-magnitude location using the Tukey biweight statistic (Beers et al. 1990) in both $BP - RP$ and G . The error in central location was estimated as $\sigma = \sqrt{v/N}$, where v is the biweight midvariance and N is the number of clump stars used for the calculation. Giant branch colors at the G magnitude of the red clump were found by overlaying an isochrone’s RGB on the data. The synthetic RGB was then shifted left and right until 50% of stars were bluer and 50%

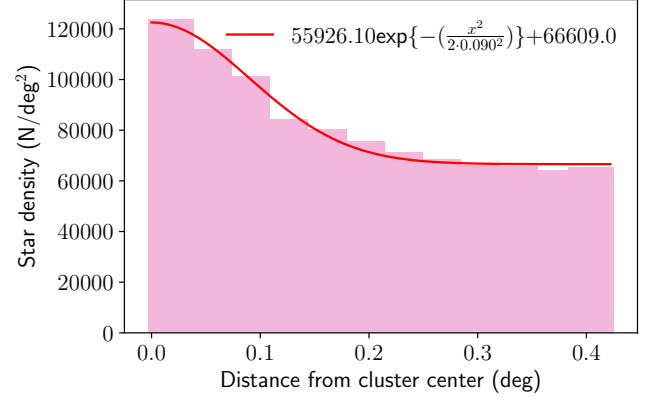


Figure 1. Stellar density in pragmatic units in annuli proceeding from the center of the sky position of NGC 7789. A Gaussian model is shown (red). Field star density is taken as the constant-density asymptote.

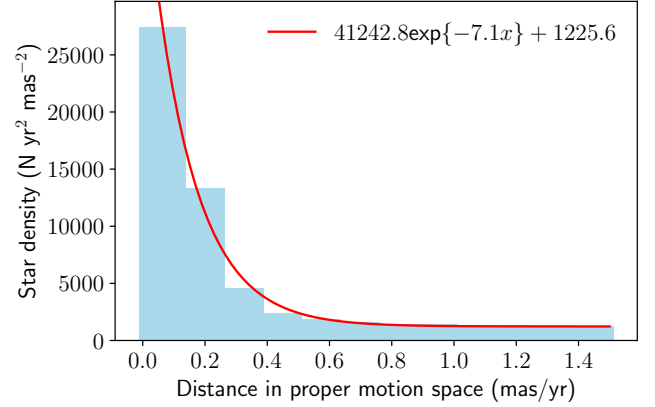


Figure 2. Stellar density in pragmatic units in annuli proceeding from the center of the proper motion location of NGC 7789 amongst the field stars. An exponential model is shown (red). Field star density is taken as the constant-density asymptote.

of stars were redder. Once placed, the $BP - RP$ magnitude at the G magnitude of the RC was read from the shifted synthetic RGB. The clump stars outnumbered the RGB stars, so we expect counting statistics on the RGB to be the largest error source. We estimated RGB placement errors by shifting the theoretical RGB $\pm$ one star from the median position, and taking the average of the absolute value of those two shifts.

We adopted ages and rough age estimate errors from Cantat-Gaudin et al. (2020), namely ± 0.175 log years for clusters younger than 9.25 log years and ± 0.15 log years for older clusters. To plot H91’s data, we use the ages reported in the paper but scale them by 0.9 to approximately account for the drift in age scale from 1991 to now. We used the algebraic formulae in Riello et al. (2021) to transform from the *Gaia* photometric system to Johnson-Cousins $B - R$. Color excess

Table 1. Age (log years) and errors from [Cantat-Gaudin et al. \(2020\)](#), color excess from [Kharchenko et al. \(2013\)](#) (magnitude), and color difference of red clump and red giant branch and errors (magnitude). Number of stars with probability greater than 0.8 and opening angle for cone search (degrees) are also included.

Name	Age (log years)	$E(BP - RP)$ (mag)	d_{BP-RP} (mag)	N_{80}	Opening angle (deg)
Pismis 18	8.76 ± 0.175	0.698	0.055 ± 0.019	321	0.099
NGC 1907	8.77 ± 0.175	0.670	0.132 ± 0.040	15	0.224
NGC 2660	8.97 ± 0.175	0.627	0.098 ± 0.008	257	0.105
Melotte 71	8.99 ± 0.175	0.139	0.125 ± 0.023	213	0.224
King 5	9.01 ± 0.175	0.897	0.087 ± 0.006	204	0.272
NGC 2477	9.05 ± 0.175	0.390	0.169 ± 0.004	1206	0.450
NGC 2236	9.06 ± 0.175	0.753	0.078 ± 0.028	139	0.210
NGC 752	9.07 ± 0.175	0.054	0.079 ± 0.005	173	1.938
NGC 1245	9.08 ± 0.175	0.335	0.103 ± 0.004	419	0.326
NGC 6940	9.13 ± 0.175	0.281	0.045 ± 0.008	447	0.100
NGC 6208	9.15 ± 0.175	0.279	0.087 ± 0.005	113	0.486
NGC 2509	9.18 ± 0.175	0.139	0.081 ± 0.012	52	0.179
NGC 7789	9.19 ± 0.175	0.296	0.124 ± 0.004	373	0.422
NGC 2506	9.22 ± 0.175	0.056	0.084 ± 0.004	1645	0.263
NGC 6939	9.23 ± 0.175	0.415	0.118 ± 0.009	364	0.245
NGC 2420	9.24 ± 0.175	0.013	0.082 ± 0.003	372	0.210
Ruprecht 68	9.26 ± 0.15	0.446	0.154 ± 0.012	57	0.221
Collinder 110	9.26 ± 0.15	0.557	0.071 ± 0.016	74	0.627
NGC 2627	9.27 ± 0.15	0.139	0.078 ± 0.007	245	0.260
Melotte 66	9.63 ± 0.15	0.167	0.127 ± 0.004	1203	0.262
NGC 2682	9.63 ± 0.15	0.067	0.075 ± 0.004	832	0.622
NGC 6791	9.80 ± 0.15	0.157	0.128 ± 0.004	3452	0.204

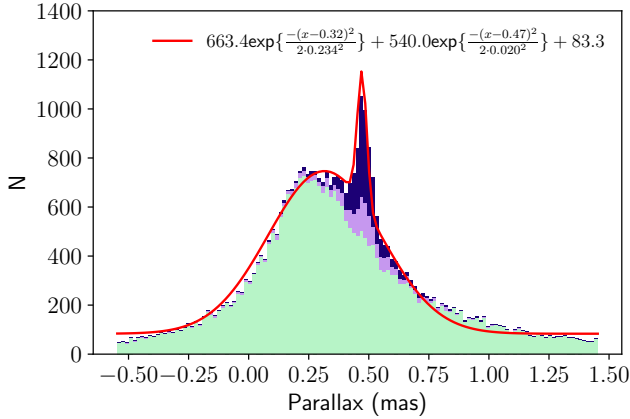


Figure 3. Distribution of stars in parallax in the NGC 7789 field. Stars nearest the proper motion locus (navy) and additional stars in a wider annulus (purple) form an obvious overdensity amongst the field stars (green). A two-Gaussian model is shown (red), with the broader component representing field stars.

from [Kharchenko et al. \(2013\)](#) was converted from $E(B-V)$ to $E(BP - RP)$ using [Casagrande & VandenBerg \(2018\)](#).

Results are summarized in Table 1 and shown in Fig. 6 with the [H91](#) fit, transformed to logarithmic age, overlaid.

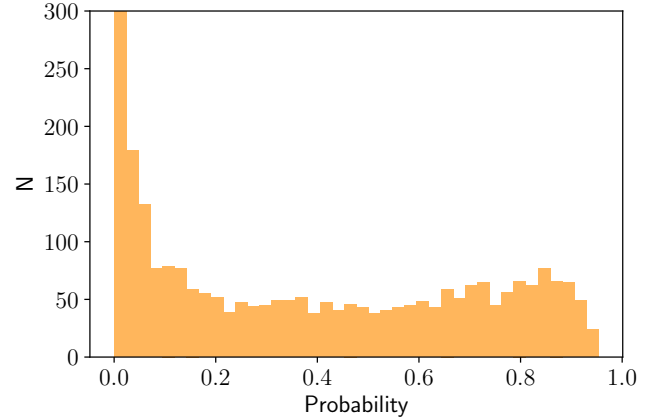


Figure 4. The resultant probability distribution after separate probabilities are combined for NGC 7789. The first bin contains about 20000 nonmember stars.

4. DISCUSSION AND CONCLUSION

From Fig. 6 we find that the [H91](#) line extrapolates poorly at the young end, and that many clusters have a larger clump-RGB color separation than the relation would predict. At $\log(\text{age}) = 9.25$, or age = 1.8 Gyr, the samples overlap in age, but every cluster in the new sample shows a wider d_{B-R}

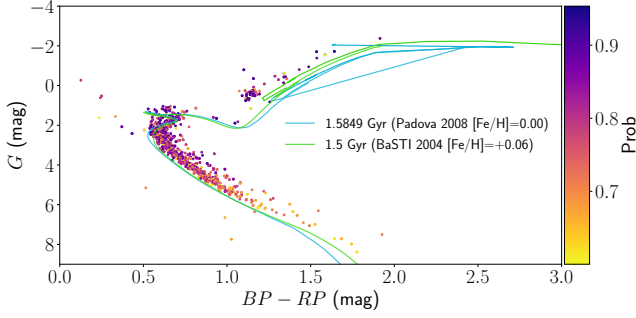


Figure 5. Color-absolute magnitude diagram in *Gaia* passbands for probable members ($P > 0.7$) of NGC 7789 using the distance and reddening from Table 1. Symbols are color mapped by membership probability as indicated by the color bar. Example isochrones from Padova (blue, with carbon-rich stars excluded) and BaSTI (green) are shown.

color separation. The magnitude of the effect exceeds any possible random or systematic uncertainties.

Even restricted to the new measurements, and further restricted to $\log \text{age} < 9.5$, the scatter appears to be astrophysical. A Kolmogorov-Smirnov test on our d_{BP-RP} values for clusters in our sample younger than 9.5 log years returned $p \approx 1.62 \times 10^{-5}$, indicating that the scatter is not drawn from a normal distribution. The cause of the scatter cannot be age, or we would see a slope in Fig. 6. We sample metallicity span less well, but within that caveat, no metallicity dependence is evident. If neither age nor metallicity is the cause of the modulation in d_{B-R} , we are driven to consider more subtle causes such as variation in helium abundance, variation in the $[\alpha/\text{Fe}]$ ratio, or variation in binary separation distributions. The investigation into the cause of the scatter must await future work.

We examine some predictions from stellar evolution theory in Fig. 7. All isochrones were translated from T_{eff} and $\log L$ to $B - R$ using the empirical transformations of Worthey & Lee (2011). We show BaSTI 5.0.1 (Pietrinferni et al. 2004), BaSTI updated (Hidalgo et al. 2018), Padova, retrieved 2011 (Marigo et al. 2008), at two abundance values, and PARSEC+COLIBRI (Bressan et al. 2012) with Riemers (Riemers 1977) $\eta = 0.2$, retrieved 2023, also at two abundance values. On the positive side, all evolutionary sets are within ~ 0.1 mag (about 150 K) of the observed color delta, and all sets show a growth in d_{B-R} for ancient stellar populations that qualitatively mimics the observations.

If precision better than 150 K are desired, theory does not match observation. In particular, reading from Fig. 7, theory predicts a metallicity dependence of roughly $\frac{d(B-R)}{d[\text{Fe}/\text{H}]} \geq -0.2$ mag per decade in heavy element abundance. A metallicity dependence this large is strictly ruled out by the data, which includes clusters from the large and small Magellanic Clouds. For example, super-metal-rich cluster NGC 6791 lies right among SMC clusters. The presence of this system-

Table 2. Integrated light systematic color error if theoretical Z dependence is allowed to flourish.

Quantity	$\log \text{age}$ = 9	$\log \text{age}$ = 9.5	$\log \text{age}$ = 10
ΔT_{eff}	172 K	93 K	212 K
$U - V$, MP, H91	0.75	1.11	1.43
$U - V$, MR, H91	1.04	1.58	1.99
$U - V$, MR, + Δ theory	1.04	1.58	1.99
Excess $U - V$	0.004	0.002	0.003
$V - K$, MP, H91	2.39	2.73	3.04
$V - K$, MR, H91	2.70	3.33	3.81
$V - K$, MR, + Δ theory	2.74	3.35	3.85
Excess $V - K$	0.038	0.021	0.037

atic among current isochrone sets affects CMD decompositions (Harris & Zaritsky 2001; Dolphin 2002). It also impacts integrated light models by introducing a fictitious color drift in the sense to make metal rich populations redder.

To estimate the size of the induced integrated light color change, we employ evolutionary population synthesis models (Worthey et al. 2022). As of this writing, these models have eight options for stellar evolution, but the earliest one (Worthey 1994) uses the H91 scheme to place the red clump. We added an extra line of code to add a temperature shift, and translated the models that span metallicity in Fig. 7 from Δcolor to $\Delta \text{temperature}$ via Worthey & Lee (2011). The results are summarized in Table 2. Because the main sequence dominates in the blue, $U - V$ is affected very little, but there is about a 4% effect in $V - K$.

Modelers should be aware of this discrepancy between current stellar evolution theory and observation because it adds to a list of ill-modeled or un-modeled effects that are far larger than the RMS fit between galaxy spectra and model integrated light spectra. Conroy et al. (2014), for example, find fits as good as 0.2% RMS, but their red clump temperatures do not adhere to H91 and so could be off by several percent. Blue stragglers are not included, which can lead to errors of over a magnitude in the terrestrial UV (Deng et al. 1999). Chemically peculiar stars, not included, alter the blue spectrum by $\sim 2\%$ in the blue (Worthey & Shi 2023). Binary evolution products, carbon stars, and metallicity-composite populations are likewise not modeled, and yet the optical spectrum matches to 0.2%. The reason the fit is excellent is a generalization the age-metallicity degeneracy (Worthey 1994), where instead of age or metallicity, one inserts the concept of stellar temperatures in general. While some astrophysical inferences, such as abundance ratios, are somewhat isolated from this degeneracy, others, such as age, metallicity, age spread, or metallicity spread are not.

Future work on red clump systematics might include cross matching *Gaia*'s star list with photometric catalogs with ac-

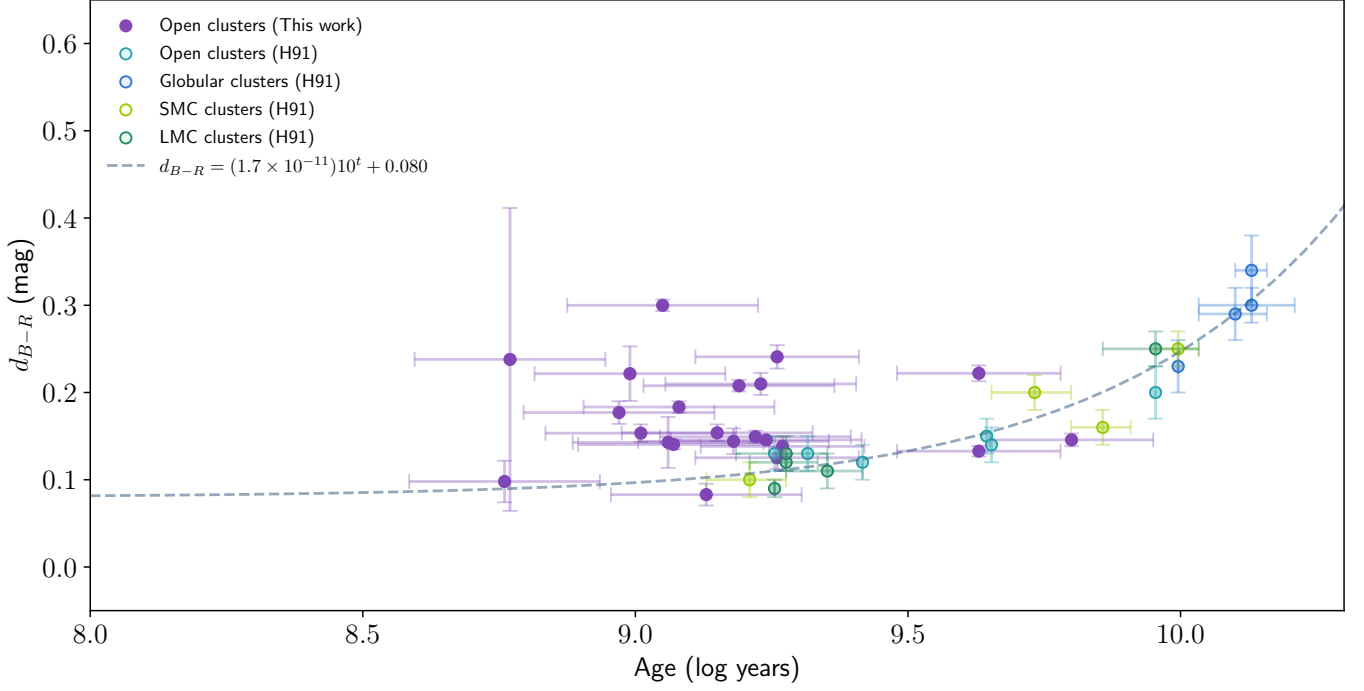


Figure 6. Color delta versus log age for open star clusters from this work (purple circles), along with open clusters (open blue-green circles), globular clusters (open blue circles), SMC clusters (open light green circles), and LMC clusters (open dark green circles) from H91. The H91 linear fit (dashed line) is also shown.

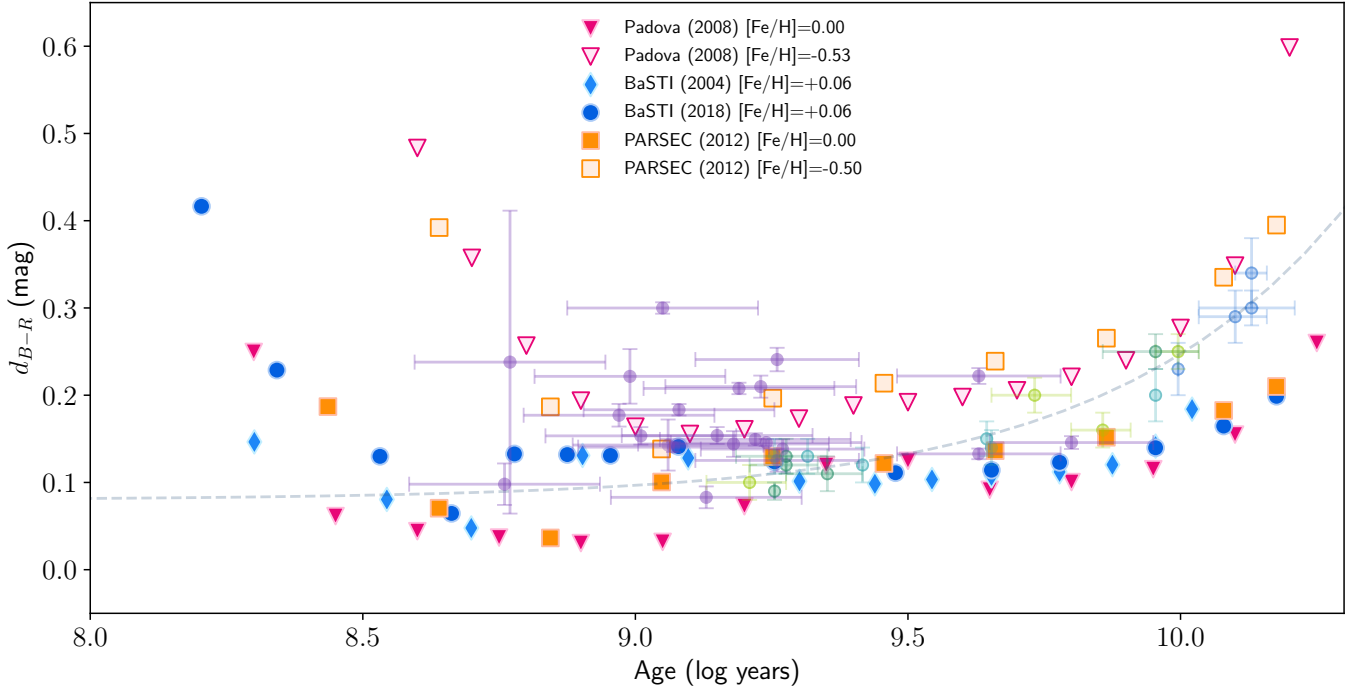


Figure 7. Color delta versus log age. Symbols are the same as Fig. 6 with the addition of predictions from various isochrone sets: We show (Marigo et al. 2008) at two abundance values (red triangles, unfilled for the metal poor sequence), BaSTI 5.0.1 (Pietrinferni et al. 2004) (blue diamonds), BaSTI (Hidalgo et al. 2018) (blue circles), and PARSEC+COLIBRI (Bressan et al. 2012) at two abundance values (orange squares, unfilled for the metal poor sequence).

curacies better than *Gaia*'s $BP - RP$. The costs might include possible loss of stars from the sample and also possible exposure to systematic error caused by transforming from heterogeneous photometric systems to a common one. To target more clusters might also be contemplated, but the key is to cover parameter space, and the present combined data of H91 and ourselves covers well what the Milky Way provides. The sample would benefit from the inclusion of SMC clusters younger than 1 Gyr. The SMC happens to have clusters aplenty at ~ 650 Myr (Glatt et al. 2010). To explore to ages much younger than that is not appropriate, and there is a hard limit at ~ 200 Myr. For ages less than ~ 200 Myr, or, equivalently, MSTO masses greater than $\sim 5 M_{\odot}$, stellar evolution through the RGB tip sparks no helium flash. The helium-burning "blue plumes" no longer resemble red clumps.

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Software: Astropy (Astropy Collaboration et al. 2013, 2018, 2022), SciPy (Virtanen et al. 2020)

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